

Momentum ETF: Backtesting, Monte Carlo, and Performance Analysis

Jake Baker

August 31, 2025

General Investment Philosophy:

The fund operates on a core philosophy of disciplined, unemotional, and systematic trend-following. We believe that human emotion, specifically fear and greed, are the greatest impediments to long-term investment success. Therefore, this strategy has been designed to remove human discretion entirely from the investment process, relying instead on a predefined, quantitative model to make all buy and sell decisions. The fund's primary objective is to participate in the significant, sustained uptrends of individual securities while seeking to preserve capital by moving to interest-bearing cash during periods of market weakness or downtrends. Unlike a traditional buy-and-hold strategy that remains fully invested through all market cycles, our approach prioritizes risk management by aiming to sidestep the most damaging bear markets. The ultimate goal is to achieve strong, risk-adjusted returns over the long term by systematically capturing the majority of major bull market trends.

Investment Methods and Rules Employed:

The core of this fund is a systematic, long-only trend-following model designed to capitalize on major uptrends in individual securities. The strategy is governed by a clear set of rules based on 50-day and 200-day simple moving averages (SMA). A long position is initiated in a security only upon a "Golden Cross" event, where its short-term trend (50-day SMA) crosses above its long-term trend (200-day SMA). This position is held for the entire duration of the established uptrend and is only exited upon a "Death Cross" event, where the 50-day SMA crosses back below the 200-day SMA. Each new position is sized to risk a modest 2% of the portfolio's currently available capital. Any cash that is not actively involved in a long position is held in a stable account compounded daily at a 2% annual risk-free interest rate to maximize total returns. The strategy does not employ short selling.

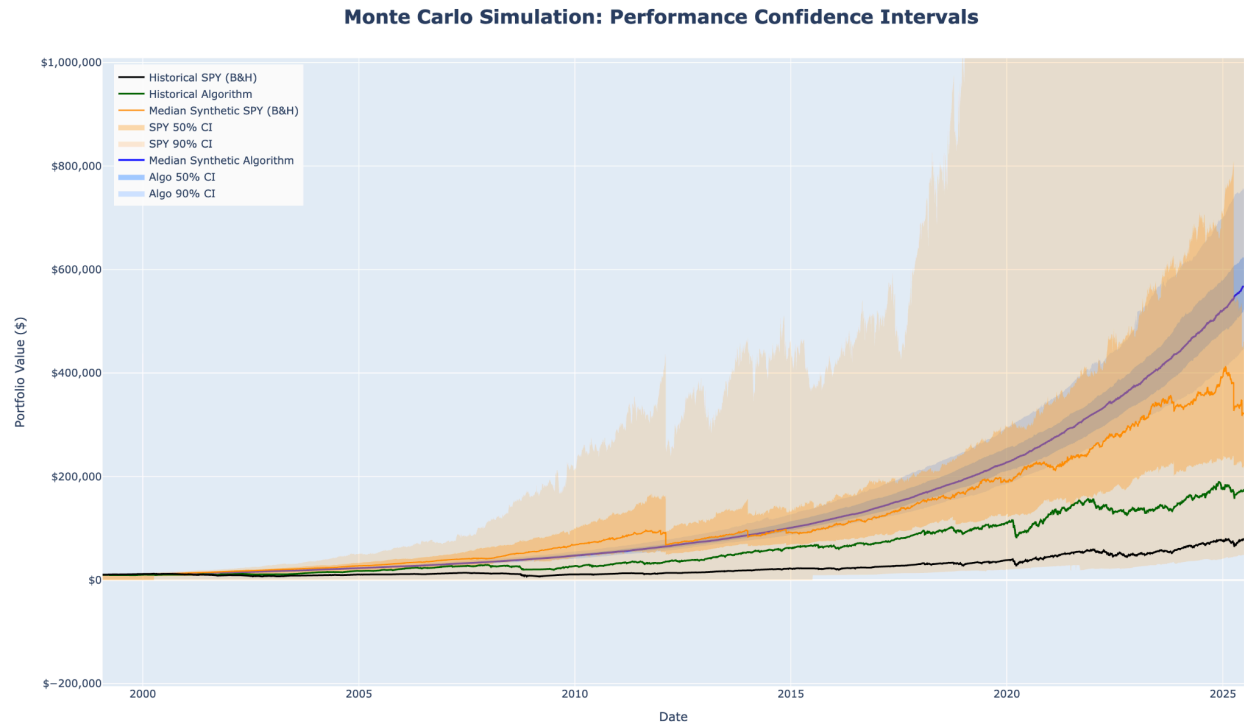
Description of Securities:

The securities used for the identification and assessment of this trading model were the roughly 500 member stocks included in the S&P 500 at this present day (August 10, 2025). For each security, the daily Open, High, Low, Close, Volume, and Log Returns were retrieved for each trading day between January 1st, 1999 and July 1st, 2025, or for as far back as the security existed on the publicly traded market. This trading window was specifically determined to include a comprehensive 25-plus year period with multiple bull runs and crashes, including the dot com bubble, the financial crisis in 2008, and the Covid crash. In addition to this, the final market capitalization for each stock was recorded on the final day of the period to assist in asset weighting for synthetically created data.

To rigorously test the trading model's robustness beyond the single path of recorded history, we employ a Monte Carlo simulation framework. This process generates hundreds of unique, realistic alternative market timelines using a technique known as block bootstrapping. For each security, its entire history of daily log returns is broken down into small, consecutive 10-day blocks. For each new simulation, these historical blocks are randomly shuffled and stitched together to create a new, full-length price history. This method is crucial as it preserves the real-world "volatility clustering"—the tendency for calm periods to follow calm periods and volatile periods to follow volatile periods—while creating a vast number of unique scenarios. From this portfolio of synthetic stocks, a corresponding synthetic benchmark index is created for each simulation run. This benchmark is a market-capitalization-weighted average of all the synthetic stocks, with the weights determined by the actual market caps recorded at the end of the data collection period. This ensures that in every simulation, the algorithm's performance is measured against a benchmark that is a true and direct representation of the specific synthetic market it is trading in.

Performance Evaluation:

The fund's strategy has demonstrated strong historical outperformance against its benchmark, even after the application of fees. Over the 25+ year backtesting period, the algorithm generated a net annualized return of 11.20%, significantly exceeding the 8.19% return of a buy-and-hold strategy on the S&P 500. This outperformance was achieved with substantially lower volatility, as evidenced by the strategy's Beta of just 0.18, indicating it is far less correlated to the market's daily fluctuations. The key measure of the strategy's edge, its net annualized Alpha, was a robust 8.09%, confirming that the returns generated were well in excess of what would be expected for the low level of risk taken. The historical performance is visually represented in the provided chart, where the algorithm's equity curve (green line) shows a smoother, more consistent upward trajectory compared to the more volatile path of the buy-and-hold benchmark (black line).



Performance Metrics Summary
(Fees: 1.0% Management & 10.0% Performance)

Metric	Historical (Gross)	Historical (Net)	Synthetic Median (Gross)	Synthetic Median (Net)
Annualized Return	11.43%	11.20%	16.50%	16.31%
Annualized Benchmark Return	8.19%	8.19%	14.03%	14.03%
Alpha (Annualized)	8.32%	8.09%	14.50%	14.31%
Beta	0.18	0.18	0.00	-0.00
Sharpe Ratio (Annualized)	0.70	0.68	9.01	7.73

To validate that this historical success was not an anomaly, the strategy was subjected to a rigorous Monte Carlo analysis of 100 randomized timelines for each stock in the S&P 500. The results confirmed the robustness of the model, with the median simulation generating a net annualized return of 16.31%, again outperforming its corresponding synthetic benchmark. The fan chart illustrates that the range of probable outcomes for the algorithm (the blue shaded area) is significantly tighter and has less downside than that of the benchmark (the orange shaded area), demonstrating the strategy's risk-managed nature. The median of these simulations produced an even stronger net annualized Alpha of 14.31% and a Beta near zero, providing compelling evidence that the strategy's edge is not dependent on a single historical path and is well-positioned to navigate a wide variety of future market conditions.

Management Recommendation:

Based on the attained metrics for the model, a financial services firm should be started around this fund. The strategy's demonstrated alpha in both historical and simulated scenarios presents a compelling case for its viability. My ideal role would be as a quantitative researcher as the developer of the model. Given the strong risk-adjusted returns, I would confidently invest my own capital, as the fund appears to be an excellent vehicle for building long-term personal wealth. The next steps are to expand the model into an actual trading bot, first executing trades in a sandbox environment, then expanding to the live market.